ST2MLE : Machine Learning for IT Engineers

Project

Machine Learning Project – Numerical and Textual Data (French Context)

Context

As part of this project, students will work on mixed data (numerical and textual) collected from French websites.
The objective is to carry out a comprehensive analysis, from data collection to modeling and interpretation, with a focus on a French economic, social, or public context.

Learning Objectives

- Master the full lifecycle of a data project (collection, cleaning, preprocessing, modeling, evaluation).
- Apply techniques for text processing and numerical data analysis.
- Explore various text vectorization techniques (BoW, TF-IDF, Doc2Vec, BERT).
- Conduct analyses and provide recommendations based on real French data.

Project Steps

1. Define a topic and identify relevant French sources.
2. Collect data (web scraping, APIs...).
3. Clean and preprocess both numerical and textual data.
4. Perform exploratory analysis and visualizations (distributions, word clouds, correlations...).
5. Apply predictive models:
    - **Numerical data**: Decision Trees, Random Forest, Boosting.
    - **Textual data**: Naive Bayes, Logistic Regression after vectorization.
6. Compare text vectorization methods: BoW, TF-IDF, Doc2Vec, BERT.
7. Provide business recommendations and submit a final report.

Technical Constraints

- Data must be exclusively from French sources.
- Texts must be in French only (use appropriate preprocessing: French lemmatization, French stopwords).
- Minimum of 1,000 data rows.
- Implementation in Python using scikit-learn, Gensim, and transformers (Hugging Face).

Deliverables

1. **Project report (.pdf)** including:
    - Description of the domain and collected data.
    - Detailed cleaning and preprocessing workflow.
    - Clear visualizations.
    - Modeling and comparisons.
    - Business recommendations.
2. **Well-documented Python code**.
3. **Oral presentation**.

Evaluation Criteria

- Quality and relevance of collected data (10%)
- Quality of data cleaning and preprocessing (10%)
- Relevance of visualizations and exploratory analysis (10%)
- Implementation of models (30%)
- Comparison and discussion of vectorization techniques (10%)
- Recommendations and critical thinking (10%)
- Quality of the report and code (5%)
- Quality of the presentation (5%)
- Q&A (10%)

Final grade: Total/5

Suggested Topics

- Analysis of job postings in France.
- Study of French customer reviews.
- Analysis of economic news articles on French companies.
- Public opinion on French public policies.
- Tourism trends in France.